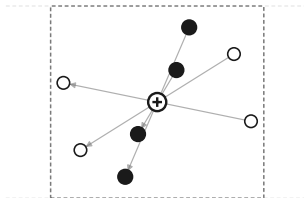


a

centrosymmetric crystal



open circle: inversion centre

arrows: $r \rightarrow -r$

$$I \cdot x = x \quad \text{and} \quad e \rightarrow -e$$

so $e = -e$, and $e = 0$

b

the only difference

O(3)

0e

1o

2e

3o

natural parity $(-1)^l$

erase the labels

SO(3)

0e

1e

2e

3e

every degree declared even

architecture, data, splits, seeds:

identical

c

true value: exactly 0

physically impossible

predicted $\|e\|_F$ (C m^{-2}) 10^0 10^{-2} 10^{-4} 10^{-6} 10^{-8}

a real piezoelectric

detection threshold

structurally zero

O(3)

SO(3)

NequIP, $n = 2,000$ crystals